

Logarithms

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Contents

1	Logarithms Worksheet	5
1.1	Logarithmic Differentiation(Revisiting Logarithms)	5

Chapter 1

Logarithms Worksheet

1.1 Logarithmic Differentiation(Revisiting Logarithms)

Let us first learn the basic definition of logarithms. First of all, we have an exponential equation of the form $a^\alpha = b$. This equation can be written in the logarithmic form as $\alpha = \log_a b$. So, we understand that logarithms is just another way of writing an equation which has exponents involved in it.

Logarithms have few basic properties :

- $\log_a p = \frac{\log_b p}{\log_b a}$ (Base Change Formula)
- $\log_a pq = \log_a p + \log_a q$
 $\log_a p^n = n \log_a p$

Now suppose, we have a function of the form,

$$y = f(x) = [u(x)]^{v(x)}.$$

By taking logarithm (to base e) the above may be rewritten as

$$\ln y = v(x) \ln[u(x)]$$

Using chain rule we may differentiate this to get

$$\frac{1}{y} \cdot \frac{dy}{dx} = v(x) \cdot \frac{1}{u(x)} \cdot u'(x) + v'(x) \ln[u(x)]$$

The main point to be noted in this method is that $f(x)$ and $u(x)$ must always be positive as otherwise their logarithms are not defined.

Q: Differentiate the following functions w.r.t. x

a) $f(x) = \sqrt{\frac{(x-3)(x^2+8)}{x^2+3x+4}}$

b) $f(x) = x^{\sin x}, x > 0$

c) $f(x) = \cos x \cdot \cos 2x \cdot \cos 3x$

d) $f(x) = (\ln x)^{\cos x}$

e) $f(x) = (\ln x)^x + x^{\ln x}$

f) $f(x) = (\sin x)^x + \sin^{-1} \sqrt{x}$

Q1: Find the domains of definition of the following functions:

a) $\log \left(\frac{x^3(x^2-4)}{(x^2-1)(x+3)} \right)$

b) $\sqrt{\log x}$

c) $\sqrt{\log_x(2-x)}$

Q2 Matrix 1: Under Column I, some equations are given. Under Column II, some solutions satisfying some of the equations are given. Match the entry in Column I with the solution satisfying it in Column II. {Note: $[]$ is the Greatest Integer Function}

Column I

(P) $|\sin^{-1} x| = \sin^{-1} x + \frac{\pi}{6}$

(Q) $\ln |x| = |\ln x|$

(R) $||x|| = [|x|]$

(S) $|x^2 - 3x + 2| > x^2 - 3x + 2$

Column II

(A) $-\frac{1}{2}$

(B) 1

(C) $\frac{3}{2}$

(D) 2

Q3 Comprehension 1: A function $f_n(x)$ is defined for all $n \in \mathbb{N}$ and $f_{n+m}(x)$ is defined as $f_{n+m}(x) = f_n(f_m(x))$ where $f_1(x) = \frac{2x-1}{x+1}$ for $x \in \mathbb{R} - \{-1\}$. Using this definition, $f_2(x) = f_1(f_1(x)) = \frac{x-1}{x}$ for $x \in \mathbb{R} - \{-1, 0\}$. Similarly $f_3(x) = f_1(f_2(x)) = \frac{2-x}{1-2x}$ for $x \in \mathbb{R} - \left\{-1, 0, \frac{1}{2}\right\}$ and so on. Based on the information, answer the questions below.

1: The domain of definition of $g(x) = |\ln(f_{34}(x))|$ is

a) $(-\infty, 1) - \left\{-1, 0, \frac{1}{2}\right\}$

b) $(-\infty, 1] - \left\{-1, 0, \frac{1}{2}\right\}$

c) $(1, \infty) - \{2\}$

d) None of these

2: The complete solution set of $|f_{71}(x)| > \left|\frac{1}{f_{73}(x)}\right|$ is

a) $(-1, 1)$

b) $\mathbb{R} - \{2\} - [-1, 1]$

c) $\mathbb{R} - \{-1, 1, 2\}$

d) None of these

3: The values of x for which $|f_{56}(x)| > f_{56}(x)$ belong to the interval

a) $\left(\frac{1}{2}, 2\right)$

b) $(2, \infty)$

c) $(0, 1)$

d) None of these

Q4: (i) Prove that for the logarithmic function $y = \ln |x|$, if the argument takes values in a G.P., then the corresponding values of the function y are in A.P.

(ii) Prove that for the exponential function $y = e^x$, if the argument takes values in a A.P., then the corresponding values of the function y are in G.P.

Index